Astronomy and physics

1. Read the article once for meaning. 2. Read it again and look at the **highlighted new words** and the **blue phrases** in context. 3. Study the tables. 4. Use at least 5 new words in speaking or writing this week.

Article: Space and the laws of the universe (156 words)

The **universe** is vast beyond imagination. Our own **solar system** is only a tiny part of a **galaxy**, and the nearest star is several **light year** distances away. Each **planet** follows its own **orbit** around the sun, held in place by **gravity**, and the **mass** of an object decides the strength of its pull. A passing **asteroid** may threaten Earth, and an **eclipse** briefly hides the sun or the moon.

Progress depends on instruments. A powerful **telescope** reveals objects too faint to see, and a **satellite** sends data about weather and communication. Any **spacecraft** sent on **space exploration** missions must reach great **velocity** to escape gravity. Scientists also study the **atom** and the **particle** that make up all matter, as well as **radiation** and **magnetic** fields. Each **experiment** helps to **rule out** wrong explanations.

Many questions remain unanswered. Researchers still **puzzle over** what makes up most of the universe, and each discovery **opens the door to** new questions.

Useful phrases from the article

PHRASE	MEANING
rule out	exclude
puzzle over	try hard to understand
opens the door to	makes possible

The 20 new words

WORD		MEANING	EXAMPLE
galaxy	n	a huge group of stars	*Our galaxy is called the Milky Way.*
orbit	n	the path of an object around another	*The Earth's orbit takes a year.*
planet	n	a large body that moves around a star	*Mars is the fourth planet from the sun.*
telescope	n	an instrument for seeing distant objects	*The telescope found a new planet.*
gravity	n	the force that pulls objects together	*Gravity keeps the moon in orbit.*
satellite	n	an object that circles a planet	*Satellites send TV signals.*
space exploration	n	the study of outer space by travel	*Space exploration costs billions.*
asteroid	n	a small rocky body in space	*An asteroid may have killed the dinosaurs.*
atom	n	the smallest part of a chemical element	*An atom has a nucleus.*

radiation	n	energy sent out as waves or particles	*Too much radiation is dangerous.*
solar system	n	the sun and all that moves around it	*There are eight planets in the solar system.*
universe	n	all of space and everything in it	*The universe is expanding.*
light year	n	the distance light travels in a year	*The star is four light years away.*
mass	n	the amount of matter in something	*The mass of the moon is smaller.*
velocity	n	speed in a given direction	*The velocity of the rocket was high.*
experiment	n	a test to learn something	*The experiment lasted three days.*
particle	n	a very small piece of matter	*Particles collide at high speed.*
magnetic	adj	able to attract metal	*The Earth has a magnetic field.*
eclipse	n	when one body hides another in space	*A solar eclipse darkens the day.*
spacecraft	n	a vehicle for travel in space	*The spacecraft landed on Mars.*

Revision: 10 words from last week

WORD		MEANING
architect	n	a person who designs buildings
facade	n	the front of a building
landmark	n	a famous building or place
sculpture	n	a work of art in stone or metal
portrait	n	a picture of a person
exhibition	n	a public show of art
masterpiece	n	an outstanding work of art
abstract	adj	not showing real objects
renovate	v	to repair and improve a building
restore	v	to bring back to its original state

Practice: fill the gaps

planet · satellite · atom · telescope · eclipse · velocity · galaxy · asteroid

1. Mars is the fourth _____ from the sun.
2. _____ send TV signals.
3. An _____ has a nucleus.
4. The _____ found a new planet.

5. A solar _____ darkens the day.

6. The _____ of the rocket was high.

7. Our _____ is called the Milky Way.

8. An _____ may have killed the dinosaurs.

More words on this topic (not in the article: choose the 10 most useful for you)

aurora (n): colored lights in the polar sky
black hole (n): a region where gravity is so strong that light cannot escape
cosmos (n): the universe
comet (n): a ball of ice moving around the sun
constellation (n): a group of stars with a name
crater (n): a bowl-shaped hole made by impact
dwarf planet (n): a small round object like Pluto
electron (n): a tiny negatively charged particle
energy (n): the power to do work
force (n): a push or pull
friction (n): resistance when things rub together
fusion (n): joining atoms together
halo (n): a circle of light
hypothesis (n): an idea to be tested
laboratory (n): a room for scientific work
lunar (adj): connected with the moon
meteor (n): a streak of light from burning rock
microscope (n): an instrument for seeing tiny things
molecule (n): a group of atoms
moon (n): a natural body that orbits a planet
nebula (n): a cloud of gas in space
neutron (n): a particle with no charge
nucleus (n): the center of an atom
observatory (n): a place for watching the stars
physicist (n): a scientist who studies physics
quantum (adj): relating to very small particles
rocket (n): a vehicle for space travel
solar flare (n): a burst of energy from the sun
supernova (n): an exploding star
weightlessness (n): the feeling of having no weight

Answers

1. planet (planet)
2. Satellites (satellite)
3. atom (atom)
4. telescope (telescope)
5. eclipse (eclipse)
6. velocity (velocity)
7. galaxy (galaxy)
8. asteroid (asteroid)